

WEB CISC3003

CHE CHI HIN
DC325182

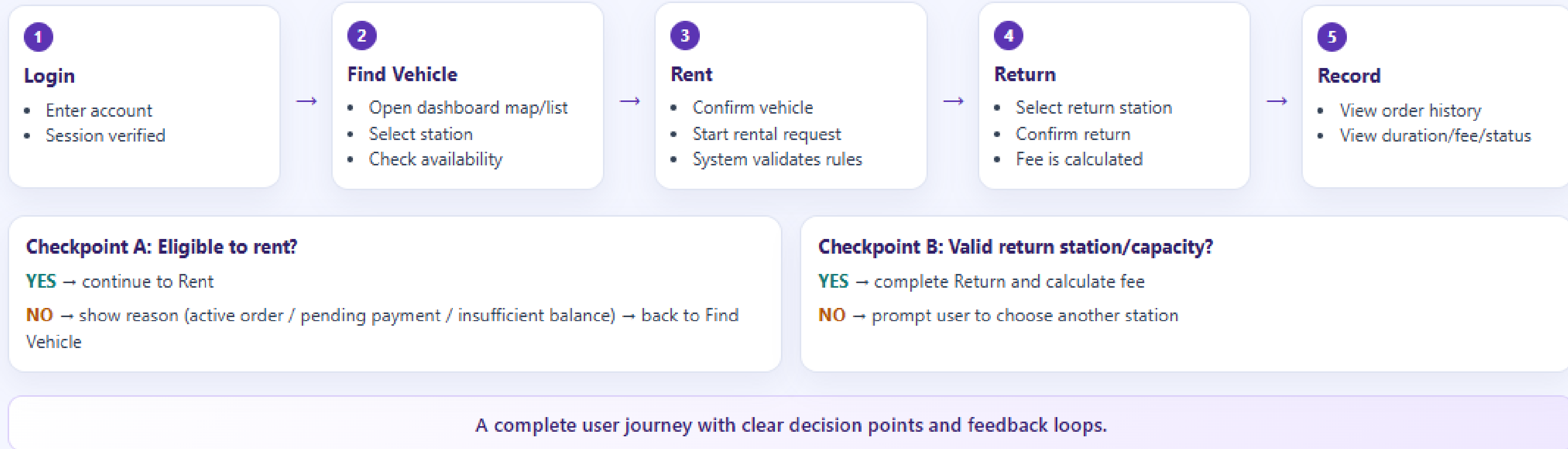


<https://cisc3003team01.is-great.net>

The interface is designed around continuous task completion, not isolated pages.

User Flow Diagram

Login → Find Vehicle → Rent → Return → Record



These principles reduce hesitation, errors, and repeated invalid actions.

Frontend Design Principles

3 Core Principles for Task-Completable User Experience

1

Next-step Clarity

DEFINITION

Users should always know what to do next.

UI EXAMPLE

Primary action buttons and flow order are visually prioritized.

USER BENEFIT

Less hesitation, faster task completion.

2

State Visibility

DEFINITION

Critical system states must be visible and easy to understand.

UI EXAMPLE

Rental status, availability, and order states are clearly displayed.

USER BENEFIT

Fewer mistakes and fewer repeated actions.

3

Actionable Feedback

DEFINITION

Feedback should tell users what happened and what to do next.

UI EXAMPLE

Clear success/failure/warning prompts with guidance.

USER BENEFIT

Higher trust and fewer invalid retries.

Good UI is not only readable - it is executable.

login.php

Users can complete authentication quickly with low cognitive load.

**UM Rental**

English ▾

Home

Services

Why us

How to use

Contact Us

Login / Use Service



Campus mobility, one account

Rent bicycles and e-scooters at UM stations—with transparent pricing and trip history.

UM Rental Login

Sign in using your Campus ID or email and password.

Continue with Google

or

Campus ID or Email

e.g. s1234567 or user@um.edu.mo

Password

Enter password

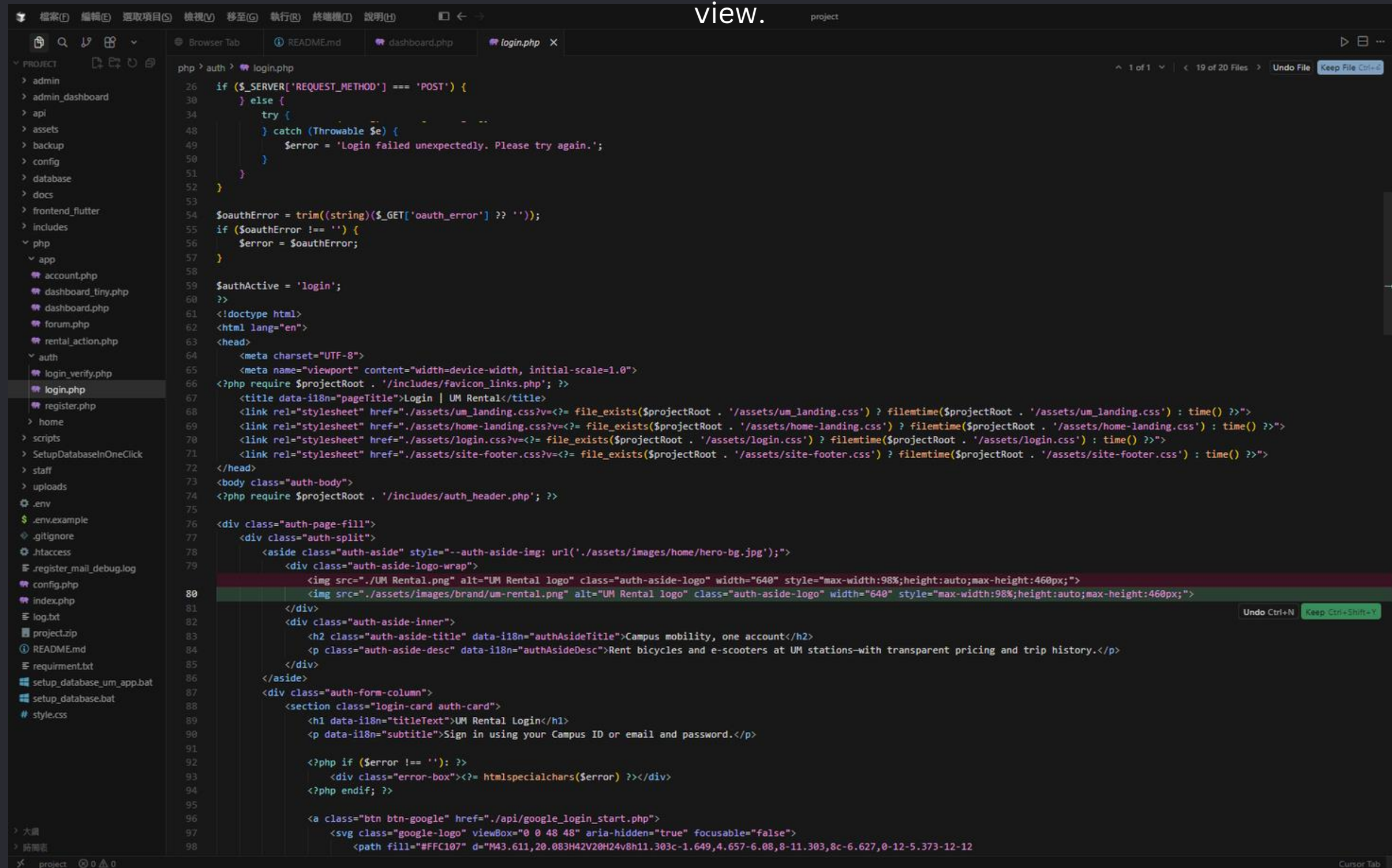
Login

New here? [Create an account](#)

[Back to Home](#)

login.php

Key operations remain visible and efficient in a rich information view.



dashboard.php

Critical actions remain reachable on small screens without layout confusion.

**UM Rental**

English

Home

Contact Us

My Account

Forum

Logout

My rental

UM RENTAL

Welcome back, Sam森

Choose where to start, pick an available bike or e-scooter, and track your trips — all in one place.

Find a ride

Pickup location

All stations

Pickup time

2026/05/10 上午 04:07

Drop-off time

2026/05/13 上午 04:07

Search

Pickup: All stations | From: 2026-05-10T04:07 | To: 2026-05-13T04:07

Campus rental stations

Each campus stop lists bicycle and e-scooter availability (including zero). Open a stop to choose a vehicle.

8 stations

Central Teaching Building (E6)

Open

Research Building (N21)

Open

Science and Technology Building (E11)

Open

Social Sciences and Humanities Building (E21)

Open

**UM Rental**

UM RENTAL

Welcome back, Sam森

Choose where to start, pick an available bike or e-scooter, and track your trips — all in one place.

Find a ride

Pickup location

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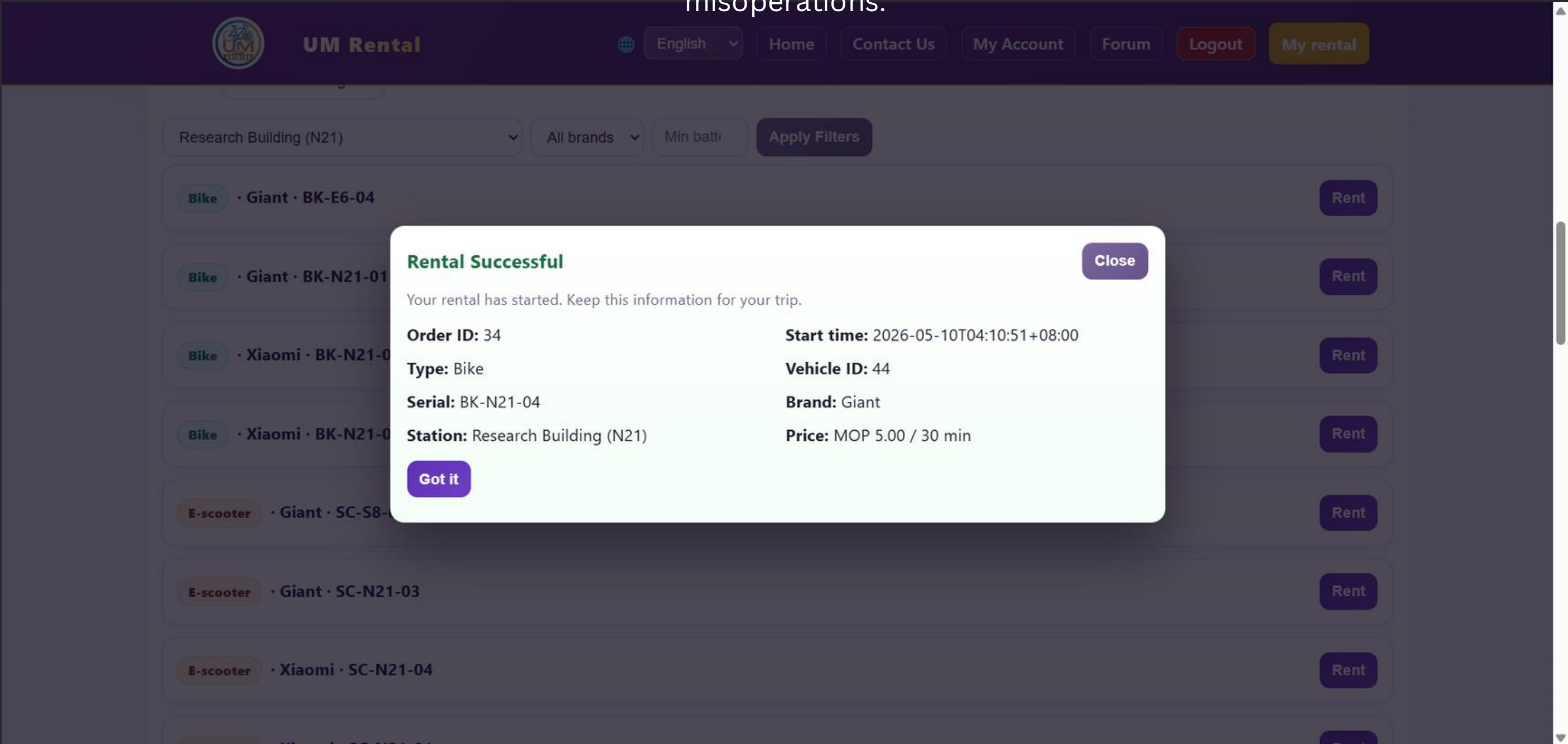
Search

dashboard.php

```
2 declare(strict_types=1);
3 $projectRoot = dirname(__DIR__, 2);
4
5 /** Log fatal's when the host hides errors (empty HTTP 500 body). Check .dashboard_fatal.log then delete it. */
6 register_shutdown_function(static function (): void {
7     $err = error_get_last();
8     if ($err === null) {
9         return;
10    }
11    if (!in_array((int) $err['type'], [E_ERROR, E_PARSE, E_CORE_ERROR, E_COMPILE_ERROR], true)) {
12        return;
13    }
14    @file_put_contents(
15        $projectRoot . '/.dashboard_fatal.log',
16        date('c') . ' ' . $err['message'] . ' @ ' . $err['file'] . ':' . $err['line'] . "\n",
17        FILE_APPEND | LOCK_EX
18    );
19 });
20
21 require_once $projectRoot . '/includes/auth.php';
22 requireUserSession();
23 startAuthSession();
24 if (!isset($_SESSION['csrf_token']) || !is_string($_SESSION['csrf_token']) || $_SESSION['csrf_token'] === '') {
25     try {
26         $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
27     } catch (Throwable $e) {
28         $_SESSION['csrf_token'] = bin2hex(hash('sha256', uniqid((string) mt_rand(), true) . microtime(true), true));
29     }
30 }
31 $csrfToken = (string)$_SESSION['csrf_token'];
32
33 $name = (string)($_SESSION['full_name'] ?? $_SESSION['campus_id'] ?? 'User');
34 $role = (string)($_SESSION['role'] ?? 'guest');
35 $activePage = 'dashboard';
36 ?>
37 <!doctype html>
38 <html lang="en">
39 <head>
40     <meta charset="UTF-8">
41     <meta name="viewport" content="width=device-width, initial-scale=1.0">
42     <?php require $projectRoot . '/includes/favicon_links.php'; ?>
43     <title>UM Rental - My rides</title>
44     <meta name="csrf-token" content="<?php echo htmlspecialchars($csrfToken) ?>">
45     <link rel="stylesheet" href="/assets/um_landing.css?v=<?php echo file_exists($projectRoot . '/assets/um_landing.css') ? filemtime($projectRoot . '/assets/um_landing.css') : time() ?>">
46     <link rel="stylesheet" href="/assets/home-landing.css?v=<?php echo file_exists($projectRoot . '/assets/home-landing.css') ? filemtime($projectRoot . '/assets/home-landing.css') : time() ?>">
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48     <link rel="stylesheet" href="/assets/site-footer.css?v=<?php echo file_exists($projectRoot . '/assets/site-footer.css') ? filemtime($projectRoot . '/assets/site-footer.css') : time() ?>">
49     <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" integrity="sha256-p4NxAoJBhIIN+hmNHrzRCf9tD/miZyoHS5obTRR9BMY=" crossorigin="anonymous">
50 </head>
51 <body class="user-dashboard" data-map-editor="<?php echo in_array($role, ['staff', 'admin'], true) ? '1' : '0' ?>">
52 <?php require $projectRoot . '/includes/public_header.php'; ?>
53
54 <main class="wrap dashboard-main">
55     <header class="dashboard-welcome">
```


dashboard.php

Consistent feedback improves trust and prevents repeated misoperations.





UM Rental

 English

Home

Contact Us

My Account

Forum

Logout

My rental

Map & stations

See where bikes and e-scooters are available around campus.

My location

Refresh GPS

Set location manually

Could not access location. Allow location in your browser to see nearby stations.



Station	Parking spots	Bikes available	E-scooters available
Central Teaching Building (E6)	22	8	5
Research Building (N21)	26	4	7
Science and Technology Building (E11)	22	4	5
Social Sciences and Humanities Building (E21)	20	6	8
Sports Complex (N8)	24	10	7
Student Activity Centre (E31)	22	4	6
University Mall (S8)	20	6	5
Wu Yee Sun Library (E2)	24	6	6

Rental Successful

Geofencing Concept

No Boundary vs With Boundary

No Boundary (Uncontrolled Scope)



- Users see irrelevant points outside service scope.
- Higher cognitive noise and slower decisions.
- More invalid exploration and confusion.

With Geofencing (Operational Scope)



- Map interactions stay within valid campus service area.
- Users make faster and more accurate decisions.
- Higher trust: visible scope matches operational scope.

● valid station/vehicle point ● irrelevant/out-of-scope point

Map Linkage Flow

Select Station -> Select Vehicle -> Confirm -> Status Update

1

Select Station

User chooses a station from map marker/list.



2

Select Vehicle

System shows station-linked available vehicles for action.



3

Confirm

User confirms rental action with clear contextual prompt.



4

Status Update

UI and backend state update consistently with feedback.

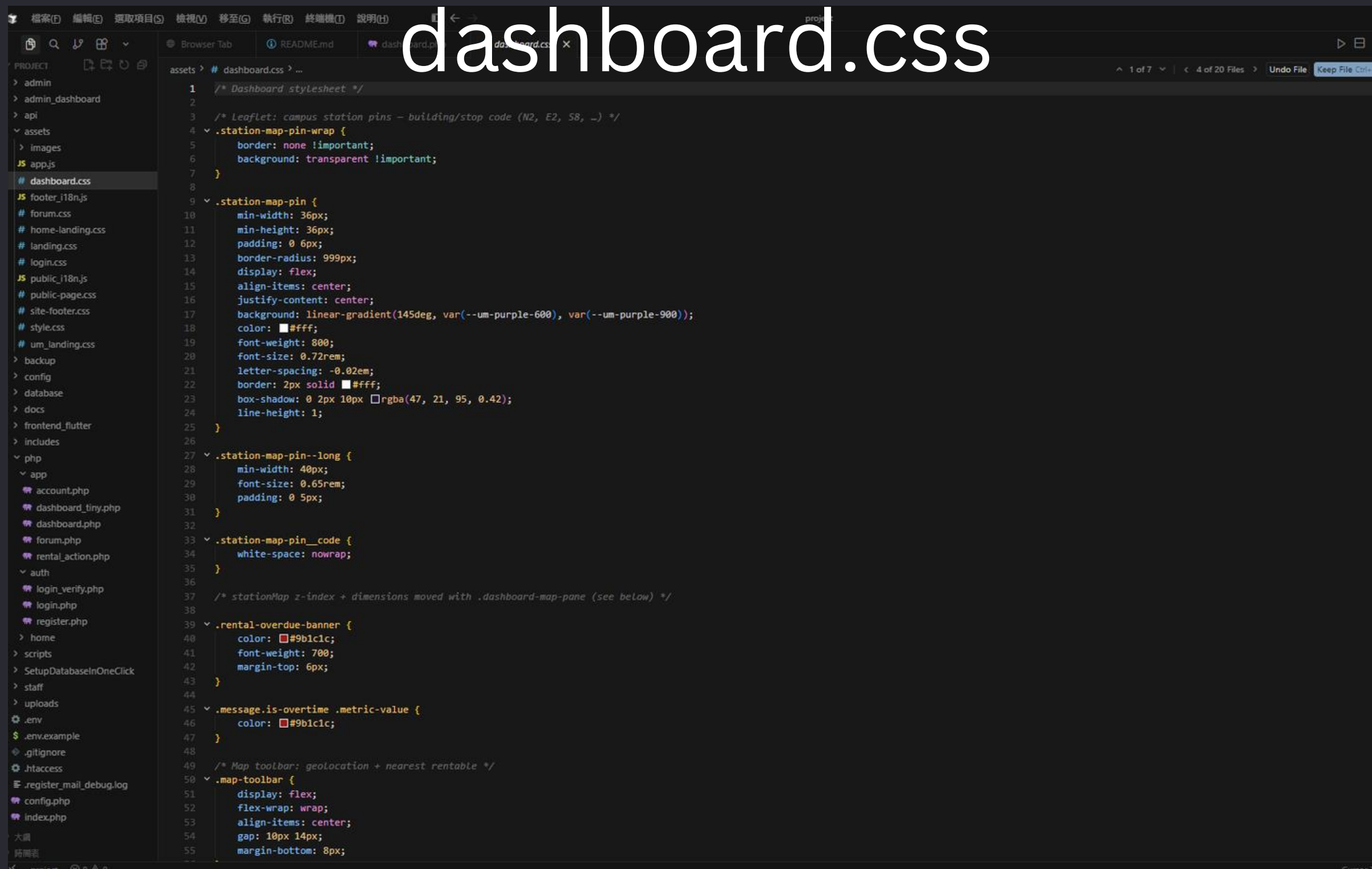
Outcome: understandable flow, lower mistakes, faster task completion.

dashboard.ph

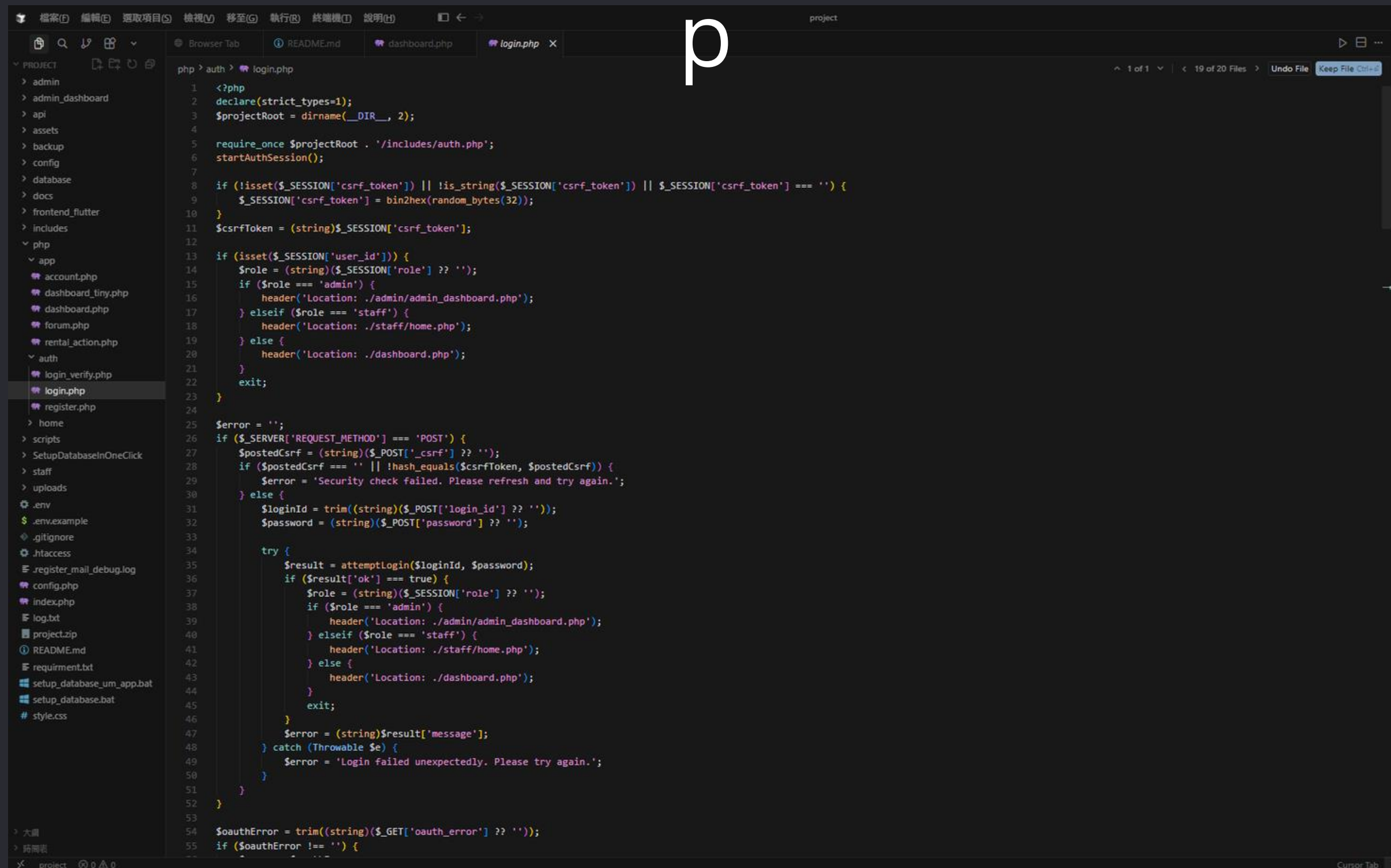
p

```
php > app > dashboard.php
1  <?php
2  declare(strict_types=1);
3  $projectRoot = dirname(__DIR__, 2);
4
5  /** Log fatals when the host hides errors (empty HTTP 500 body). Check .dashboard_fatal.log then delete it. */
6  register_shutdown_function(static function (): void {
7      $serr = error_get_last();
8      if ($serr === null) {
9          return;
10     }
11     if (!in_array((int) $serr['type'], [E_ERROR, E_PARSE, E_CORE_ERROR, E_COMPILE_ERROR], true)) {
12         return;
13     }
14     @file_put_contents(
15         $projectRoot . '/.dashboard_fatal.log',
16         date('c') . ' ' . $serr['message'] . ' @ ' . $serr['file'] . ':' . $serr['line'] . "\n",
17         FILE_APPEND | LOCK_EX
18     );
19 });
20
21 require_once $projectRoot . '/includes/auth.php';
22 requireUserSession();
23 startAuthSession();
24 if (!isset($_SESSION['csrf_token']) || !is_string($_SESSION['csrf_token']) || $_SESSION['csrf_token'] === '') {
25     try {
26         $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
27     } catch (Throwable $e) {
28         $_SESSION['csrf_token'] = bin2hex(hash('sha256', uniqid((string) mt_rand(), true) . microtime(true), true));
29     }
30 }
31 $csrfToken = (string) $_SESSION['csrf_token'];
32
33 $name = (string) ($_SESSION['full_name'] ?? $_SESSION['campus_id'] ?? 'User');
34 $role = (string) ($_SESSION['role'] ?? 'guest');
35 $activePage = 'dashboard';
36 ??
37 <!doctype html>
38 <html lang="en">
39 <head>
40     <meta charset="UTF-8">
41     <meta name="viewport" content="width=device-width, initial-scale=1.0">
42     <?php require $projectRoot . '/includes/favicon_links.php'; ??
43     <title>UM Rental - My rides</title>
44     <meta name="csrf-token" content="<?php echo htmlspecialchars($csrfToken) ?? ">
45     <link rel="stylesheet" href="/assets/um_landing.css?v=<?php echo file_exists($projectRoot . '/assets/um_landing.css') ? filemtime($projectRoot . '/assets/um_landing.css') : time() ?? ">
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49     <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" integrity="sha256-p4NxAoJBhIIN+hmMhrzRCf9tD/miZyoHS5obTRR98MY=" crossorigin="anonymous">
50 </head>
51 <body class="user-dashboard" data-map-editor="<?php echo in_array($role, ['staff', 'admin'], true) ? '1' : '0' ?? ">
52 <?php require $projectRoot . '/includes/public_header.php'; ??
53
54 <main class="wrap dashboard-main">
55     <header class="dashboard-welcome">
```


dashboard.css



login.php



```
1 <?php
2 declare(strict_types=1);
3 $projectRoot = dirname(__DIR__, 2);
4
5 require_once $projectRoot . '/includes/auth.php';
6 startAuthSession();
7
8 if (!isset($_SESSION['csrf_token']) || !is_string($_SESSION['csrf_token']) || $_SESSION['csrf_token'] === '') {
9     $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
10 }
11 $csrfToken = (string)$_SESSION['csrf_token'];
12
13 if (isset($_SESSION['user_id'])) {
14     $role = (string)$_SESSION['role'] ?? '';
15     if ($role === 'admin') {
16         header('Location: ./admin/admin_dashboard.php');
17     } elseif ($role === 'staff') {
18         header('Location: ./staff/home.php');
19     } else {
20         header('Location: ./dashboard.php');
21     }
22     exit;
23 }
24
25 $error = '';
26 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
27     $postedCsrf = (string)$_POST['_csrf'] ?? '';
28     if ($postedCsrf === '' || !hash_equals($csrfToken, $postedCsrf)) {
29         $error = 'Security check failed. Please refresh and try again.';
30     } else {
31         $loginId = trim((string)$_POST['login_id'] ?? '');
32         $password = (string)$_POST['password'] ?? '';
33
34         try {
35             $result = attemptLogin($loginId, $password);
36             if ($result['ok'] === true) {
37                 $role = (string)$_SESSION['role'] ?? '';
38                 if ($role === 'admin') {
39                     header('Location: ./admin/admin_dashboard.php');
40                 } elseif ($role === 'staff') {
41                     header('Location: ./staff/home.php');
42                 } else {
43                     header('Location: ./dashboard.php');
44                 }
45                 exit;
46             }
47             $error = (string)$result['message'];
48         } catch (Throwable $e) {
49             $error = 'Login failed unexpectedly. Please try again.';
50         }
51     }
52 }
53
54 $oauthError = trim((string)$_GET['oauth_error'] ?? '');
55 if ($oauthError !== '') {
56     // ...
57 }
```


register.ph

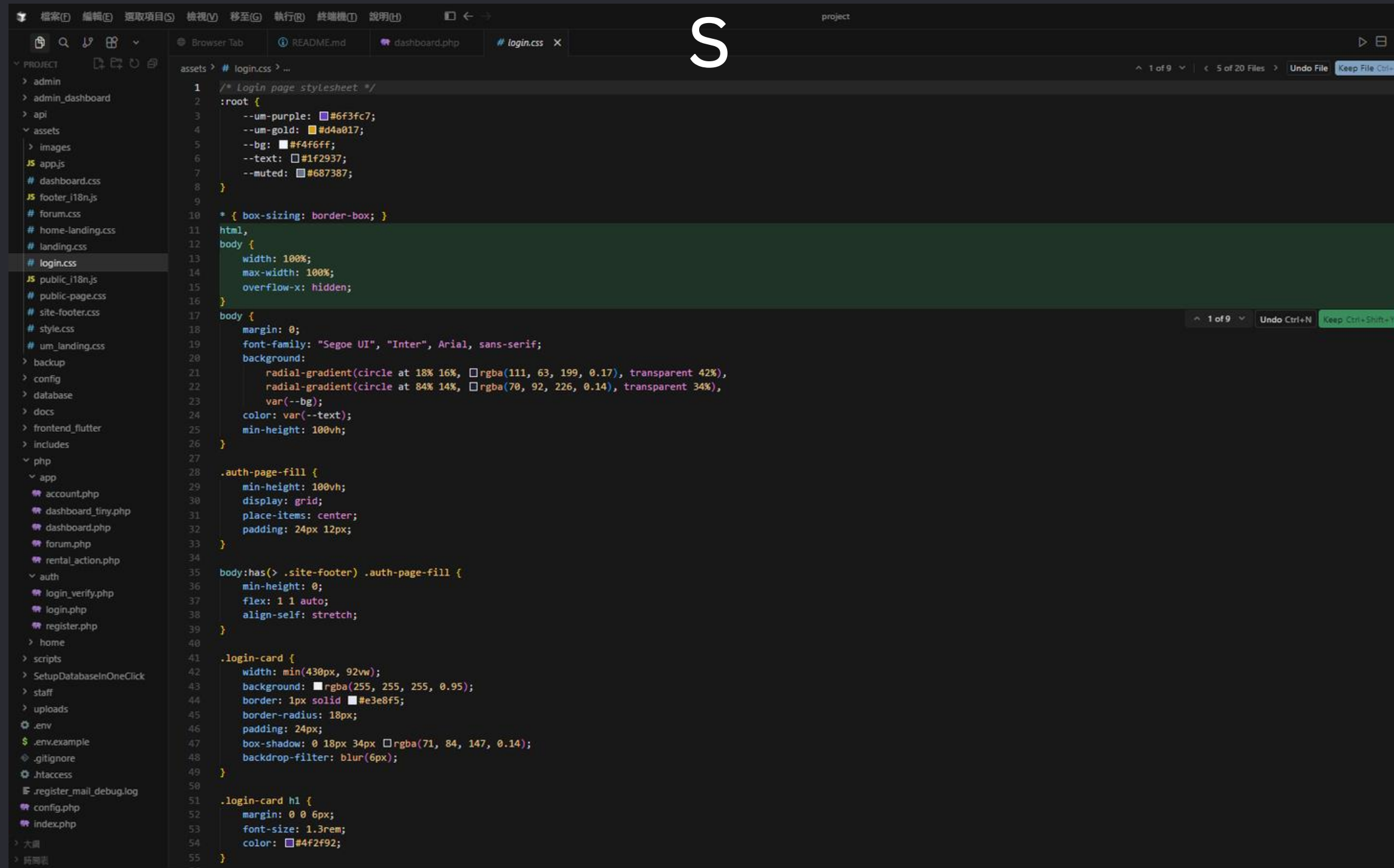
p

```
PROJECT
├── admin
├── admin_dashboard
├── api
├── assets
├── backup
├── config
├── database
├── docs
├── frontend_flutter
├── includes
├── php
│   ├── app
│   │   ├── account.php
│   │   ├── dashboard_tiny.php
│   │   ├── dashboard.php
│   │   ├── forum.php
│   │   ├── rental_action.php
│   │   └── auth
│   │       ├── login_verify.php
│   │       ├── login.php
│   │       └── register.php
│   ├── home
│   ├── scripts
│   ├── SetupDatabaseOnClick
│   ├── staff
│   ├── uploads
│   ├── .env
│   ├── .env.example
│   ├── .gitignore
│   ├── .htaccess
│   ├── .register_mail_debug.log
│   ├── config.php
│   ├── index.php
│   ├── log.txt
│   ├── project.zip
│   ├── README.md
│   ├── requirement.txt
│   ├── setup_database_um_app.bat
│   ├── setup_database.bat
│   └── style.css
├── 大綱
└── 路徑表
```

```
php > auth > register.php
9  <head>
10  <meta charset="UTF-8">
11  <meta name="viewport" content="width=device-width, initial-scale=1.0">
12  <?php require $projectRoot . '/includes/favicon_links.php'; ?>
13  <title data-i18n="pageTitle">Register | UM Rental</title>
14  <link rel="stylesheet" href="/assets/um_landing.css?v=<? file_exists($projectRoot . '/assets/um_landing.css') ? filemtime($projectRoot . '/assets/um_landing.css') : time() ?>">
15  <link rel="stylesheet" href="/assets/home-landing.css?v=<? file_exists($projectRoot . '/assets/home-landing.css') ? filemtime($projectRoot . '/assets/home-landing.css') : time() ?>">
16  <link rel="stylesheet" href="/assets/login.css?v=<? file_exists($projectRoot . '/assets/login.css') ? filemtime($projectRoot . '/assets/login.css') : time() ?>">
17  <link rel="stylesheet" href="/assets/site-footer.css?v=<? file_exists($projectRoot . '/assets/site-footer.css') ? filemtime($projectRoot . '/assets/site-footer.css') : time() ?>">
18  </head>
19  <body class="auth-body">
20  <?php require $projectRoot . '/includes/auth_header.php'; ?>
21
22  <div class="auth-page-fill">
23  <div class="auth-split">
24  <aside class="auth-aside" style="--auth-aside-img: url('/assets/images/home/feature-campus.jpg');">
25  <div class="auth-aside-logo-wrap">
26  
27  
28  </div>
29  <div class="auth-aside-inner">
30  <h2 class="auth-aside-title" data-i18n="authAsideTitleReg">Create your campus profile</h2>
31  <p class="auth-aside-desc" data-i18n="authAsideDescReg">Register with your UM email, then sign in with Campus ID or email.</p>
32  </div>
33  </aside>
34  <div class="auth-form-column">
35  <section class="login-card auth-card auth-card-wide">
36  <h1 data-i18n="title">Create Account</h1>
37  <p data-i18n="subtitle">Register with your email, then log in using Campus ID or email.</p>
38
39  <div id="messageBox" class="error-box is-hidden"></div>
40
41  <form id="registerForm">
42  <label for="campus_id" data-i18n="campusId">Campus ID</label>
43  <input id="campus_id" required placeholder="e.g. s1234567" data-i18n-placeholder="campusIdPlaceholder">
44
45  <label for="full_name" data-i18n="fullName">Full Name</label>
46  <input id="full_name" required placeholder="e.g. Chan Tai Man" data-i18n-placeholder="fullNamePlaceholder">
47
48  <label for="role" data-i18n="roleLabel">Account type</label>
49  <select id="role" required>
50  <option value="student" data-i18n="roleStudent">Student</option>
51  <option value="teacher" data-i18n="roleTeacher">Teacher / faculty</option>
52  </select>
53
54  <label for="email" data-i18n="email">Email</label>
55  <input id="email" type="email" required placeholder="e.g. user@um.edu.mo" data-i18n-placeholder="emailPlaceholder">
56
57  <div class="verification-row">
58  <div class="verification-input-wrap">
59  <label for="verification_code" data-i18n="verificationCode">Verification code</label>
60  <input id="verification_code" inputmode="numeric" maxlength="6" required placeholder="6-digit code" data-i18n-placeholder="verificationCodePlaceholder">
61  </div>
62  <button class="btn verification-send-btn" type="button" id="sendCodeBtn" data-i18n="sendCodeBtn">Send Code</button>
63  </div>
```

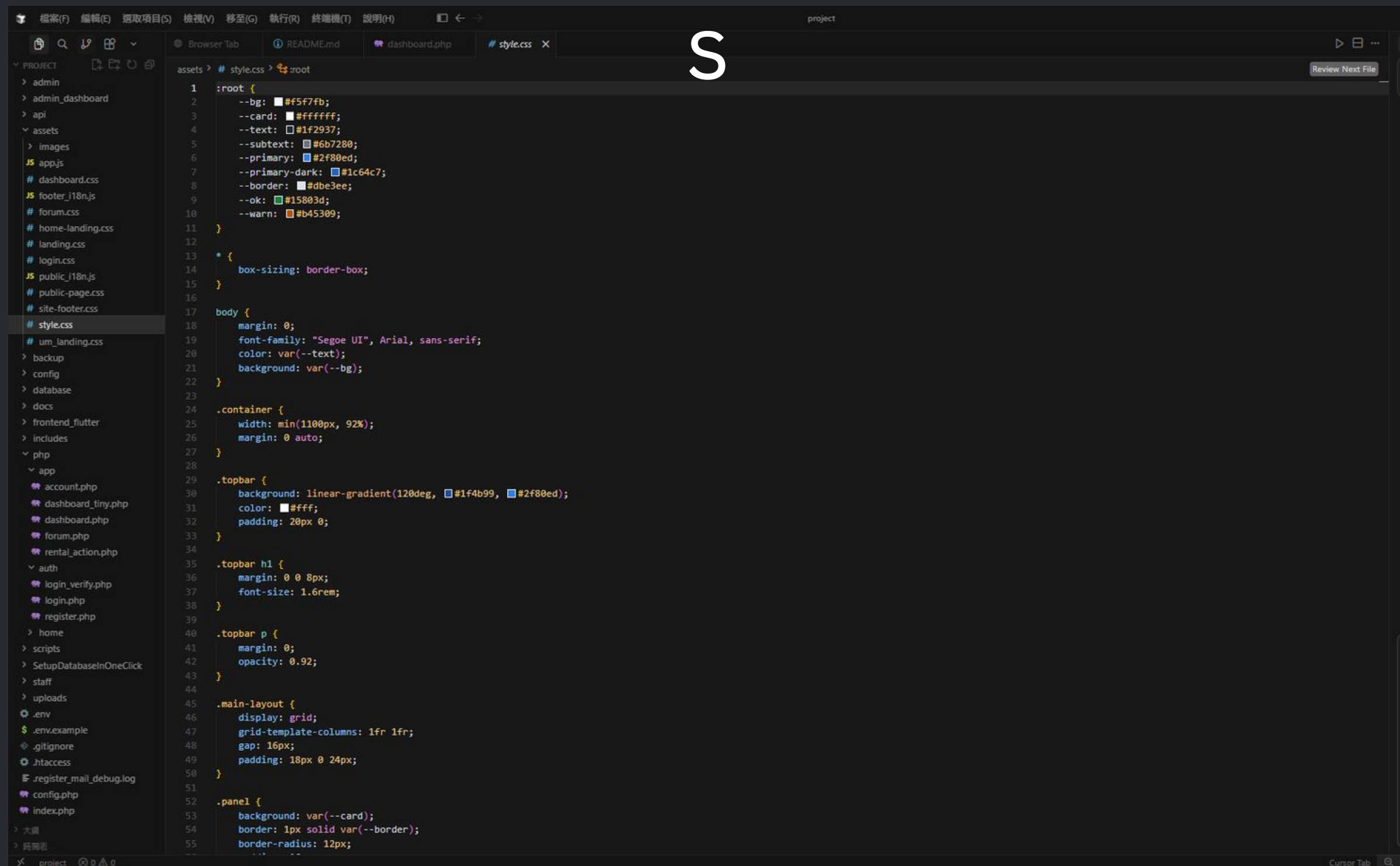
login.cs

S



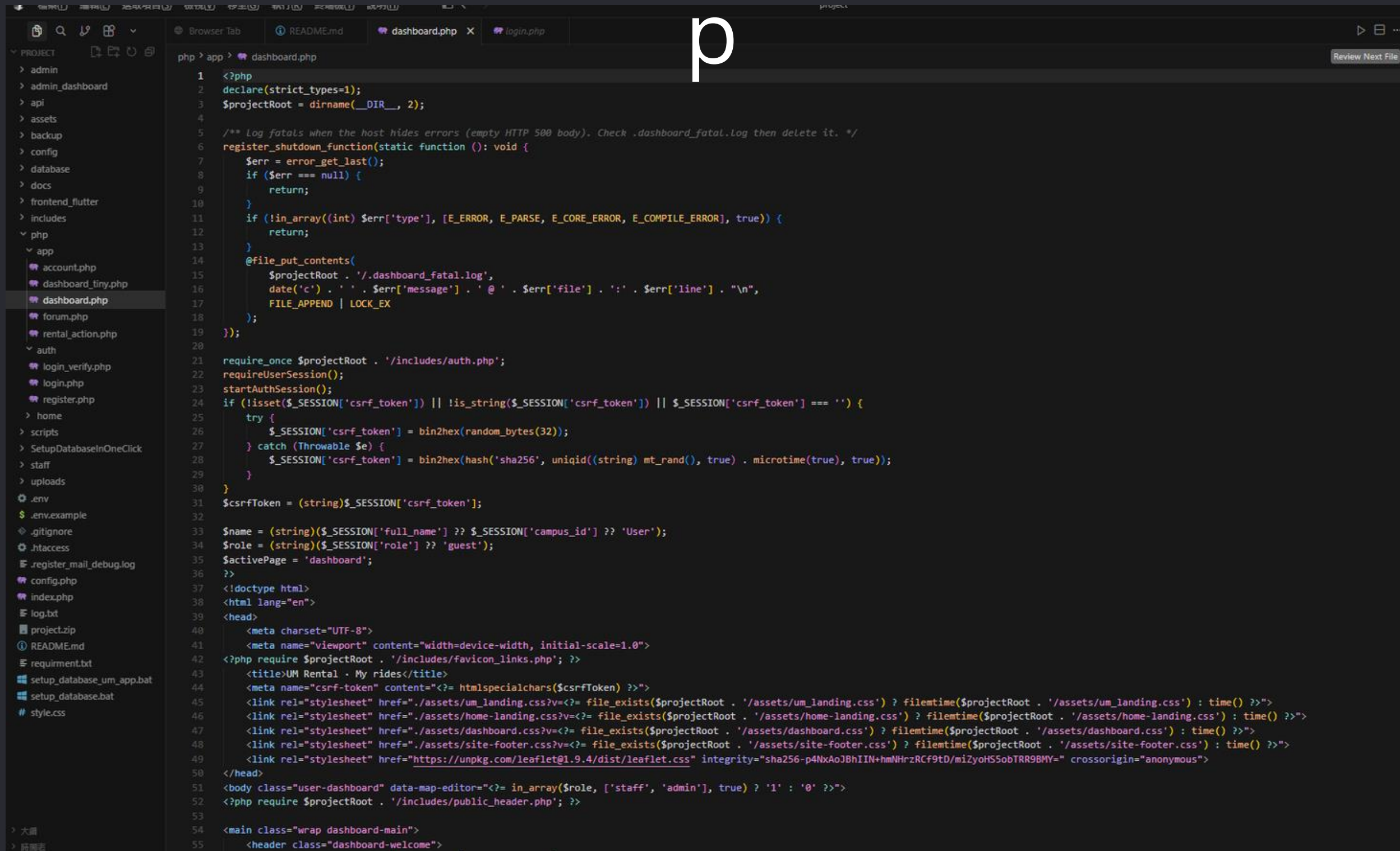
style.cs

S



dashboard.ph

p



```
1 <?php
2 declare(strict_types=1);
3 $projectRoot = dirname(__DIR__, 2);
4
5 /** Log fatals when the host hides errors (empty HTTP 500 body). Check .dashboard_fatal.log then delete it. */
6 register_shutdown_function(static function (): void {
7     $serr = error_get_last();
8     if ($serr === null) {
9         return;
10    }
11    if (!in_array((int) $serr['type'], [E_ERROR, E_PARSE, E_CORE_ERROR, E_COMPILE_ERROR], true)) {
12        return;
13    }
14    @file_put_contents(
15        $projectRoot . '/.dashboard_fatal.log',
16        date('c') . ' ' . $serr['message'] . ' @ ' . $serr['file'] . ':' . $serr['line'] . "\n",
17        FILE_APPEND | LOCK_EX
18    );
19 });
20
21 require_once $projectRoot . '/includes/auth.php';
22 requireUserSession();
23 startAuthSession();
24 if (!isset($_SESSION['csrf_token']) || !is_string($_SESSION['csrf_token']) || $_SESSION['csrf_token'] === '') {
25     try {
26         $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
27     } catch (Throwable $e) {
28         $_SESSION['csrf_token'] = bin2hex(hash('sha256', uniqid((string) mt_rand(), true) . microtime(true), true));
29     }
30 }
31 $csrfToken = (string) $_SESSION['csrf_token'];
32
33 $name = (string) ($_SESSION['full_name'] ?? $_SESSION['campus_id'] ?? 'User');
34 $role = (string) ($_SESSION['role'] ?? 'guest');
35 $activePage = 'dashboard';
36 ??
37 <!doctype html>
38 <html lang="en">
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41     <meta name="viewport" content="width=device-width, initial-scale=1.0">
42     <?php require $projectRoot . '/includes/favicon_links.php'; ??
43     <title>UM Rental - My rides</title>
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45     <link rel="stylesheet" href="/assets/um_landing.css?v=<?php echo file_exists($projectRoot . '/assets/um_landing.css') ? filemtime($projectRoot . '/assets/um_landing.css') : time() ?>">
46     <link rel="stylesheet" href="/assets/home-landing.css?v=<?php echo file_exists($projectRoot . '/assets/home-landing.css') ? filemtime($projectRoot . '/assets/home-landing.css') : time() ?>">
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50 </head>
51 <body class="user-dashboard" data-map-editor="<?php echo in_array($role, ['staff', 'admin'], true) ? '1' : '0' ?>">
52 <?php require $projectRoot . '/includes/public_header.php'; ??
53
54 <main class="wrap dashboard-main">
55     <header class="dashboard-welcome">
```

stations.ph

p

```
1 <?php
2 declare(strict_types=1);
3
4 header('Content-Type: application/json; charset=utf-8');
5 require_once __DIR__ . '/../config/db_connect.php';
6 require_once __DIR__ . '/../includes/rental_station_locale.php';
7
8 /**
9  * 站點與可用車輛查詢 API
10  * 支援篩選：
11  * - stationID
12  * - vehicleType (bicycle / scooter)
13  * - brand
14  */
15
16 function respond(int $statusCode, array $payload): void
17 {
18     http_response_code($statusCode);
19     echo json_encode($payload, JSON_UNESCAPED_UNICODE);
20     exit;
21 }
22
23 if ($_SERVER['REQUEST_METHOD'] !== 'GET') {
24     respond(405, ['success' => false, 'message' => '只允許 GET 請求']);
25 }
26
27 $stationID = isset($_GET['stationID']) ? (int)$_GET['stationID'] : null;
28 $vehicleType = isset($_GET['vehicleType']) ? trim((string)$_GET['vehicleType']) : null;
29 $brand = isset($_GET['brand']) ? trim((string)$_GET['brand']) : null;
30
31 $allowedTypes = ['bicycle', 'scooter'];
32 if ($vehicleType !== null && $vehicleType !== '' && !in_array($vehicleType, $allowedTypes, true)) {
33     respond(422, ['success' => false, 'message' => 'vehicleType 只能是 bicycle 或 scooter']);
34 }
35
36 try {
37     $hasLocale = rentalStationsHasLocaleColumns($pdo);
38
39     $sql = $hasLocale
40         ? '
41         SELECT
42             rs.id,
43             rs.name,
44             rs.name_zh_cn,
45             rs.name_zh_tw,
46             rs.latitude,
47             rs.longitude,
48             rs.capacity,
49             SUM(CASE WHEN v.status = "available" THEN 1 ELSE 0 END) AS availableVehicles,
50             COUNT(v.id) AS totalVehicles
51         FROM rental_stations rs
52         LEFT JOIN vehicles v ON v.station_id = rs.id
53         LEFT JOIN vehicle_types vt ON v.type_id = vt.id
54         LEFT JOIN brands b ON v.brand_id = b.id
55         WHERE rs.status = "active"
56     '
```


dashboard_api.ph

project

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api > dashboard_api.php

```
15 if (!isset($_SESSION['user_id'])) {
16     http_response_code(401);
17     echo json_encode(['success' => false, 'message' => 'Unauthorized']);
18     exit;
19 }
20
21 if (!roleCanRent((string)($_SESSION['role'] ?? ''))) {
22     http_response_code(403);
23     echo json_encode(['success' => false, 'message' => 'This account cannot use the rental service.', 'message_key' => 'errRentalNotAllowed']);
24     exit;
25 }
26
27 $userId = (int)$_SESSION['user_id'];
28 $action = $_GET['action'] ?? '';
29 $method = $_SERVER['REQUEST_METHOD'] ?? 'GET';
30 $pdo = db();
31 if (in_array($method, ['POST', 'PUT', 'PATCH', 'DELETE'], true)) {
32     verifyCsrfOrFail();
33     rateLimit('dashboard_api:' . $action . ':user:' . $userId . ':ip:' . clientIp(), 120, 300, 120);
34 }
35
36 /**
37  * Reads JSON body safely for POST operations.
38  */
39 function readBody(): array
40 {
41     $raw = file_get_contents('php://input');
42     $data = json_decode($raw, true);
43     return is_array($data) ? $data : [];
44 }
45
46 /**
47  * Writes JSON response with status code.
48  */
49 function respond(int $code, array $payload): void
50 {
51     http_response_code($code);
52     echo json_encode($payload);
53     exit;
54 }
55
56 /** Same bounds as dashboard Leaflet UM_BOUNDS (must stay in sync). */
57 function umCampusCoordsOk(float $lat, float $lng): bool
58 {
59     return $lat >= 22.1150 && $lat <= 22.1455 && $lng >= 113.5250 && $lng <= 113.5600;
60 }
61
62 if ($action === 'get_dashboard' && $method === 'GET') {
63     try {
64         $stationId = (int)($_GET['stationID'] ?? 0);
65         $brandId = (int)($_GET['brandID'] ?? 0);
66         $minBattery = (int)($_GET['minBattery'] ?? 0);
67         $bikePage = max(1, (int)($_GET['bikePage'] ?? 1));
68         $bikePerPage = max(5, min(50, (int)($_GET['bikePerPage'] ?? 10)));
69         $orderPage = max(1, (int)($_GET['orderPage'] ?? 1));
```